VALSE 2026

视觉与学习青年学者研讨会

Wechat Github

Uniform DiscRete DiffuSion with Metric PAth for Video Generation

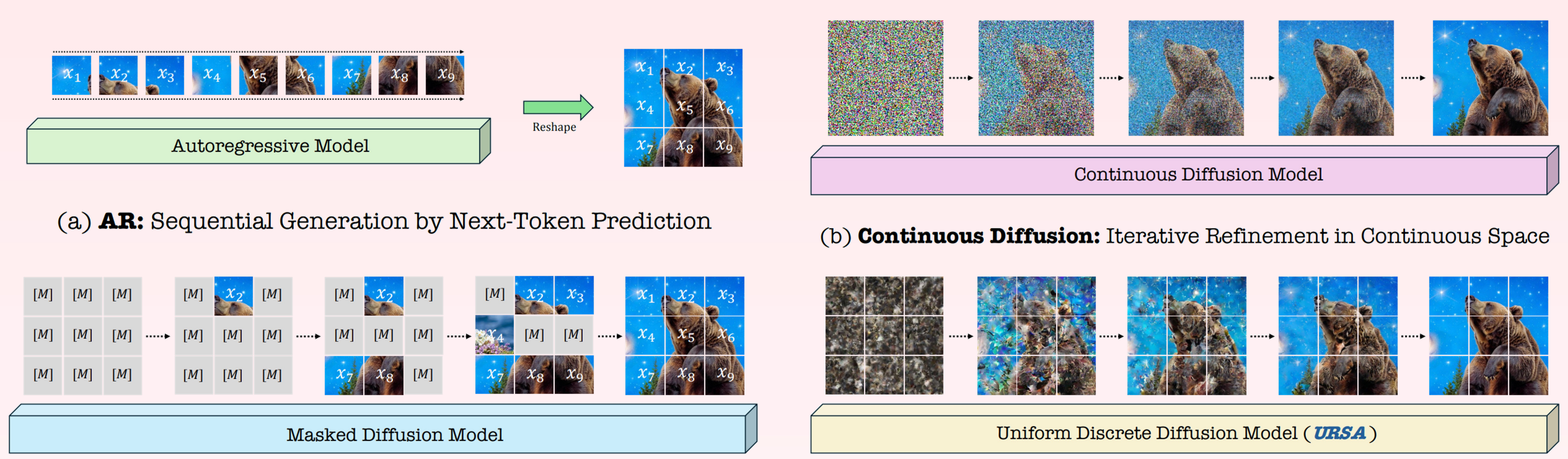
Haoge Deng[1,4*], Ting Pan[2,4*], Fan Zhang[4*], Liu Yang[3,4*], Zhuoyan Luo[4], Yufeng Cui[4] , Wenxuan Wang[4]
Chunhua Shen[3], Shiguang Shan[2], Zhaoxiang Zhang[1†] , Xinlong Wang[4†]

* Equal contribution † Corresponding Author

1 National Laboratory of Pattern Recognition, CASIA 2 Key Laboratory of Intelligent Information Processing, CASICT 3 Zhejiang University 4 Beijing Academy of Artificial Intelligence

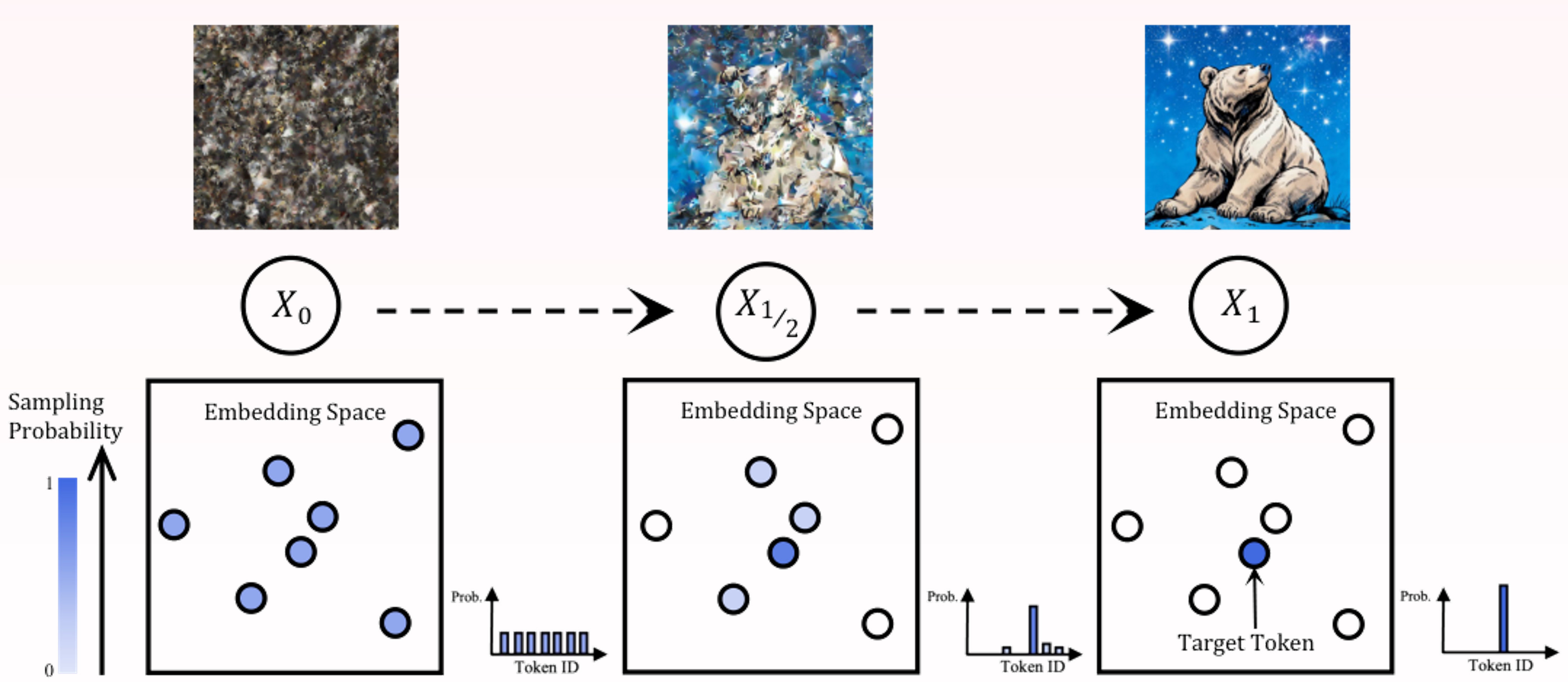
Introduction

Unlike classic AR models and MDM that adopt non-refinable local generation, where produced tokens are fixed once generated, URSA emphasizes ***iterative refinement*** over global discrete tokens, conceptually aligning discrete methods with continuous counterparts, and substantially narrowing their performance gap.



Methods

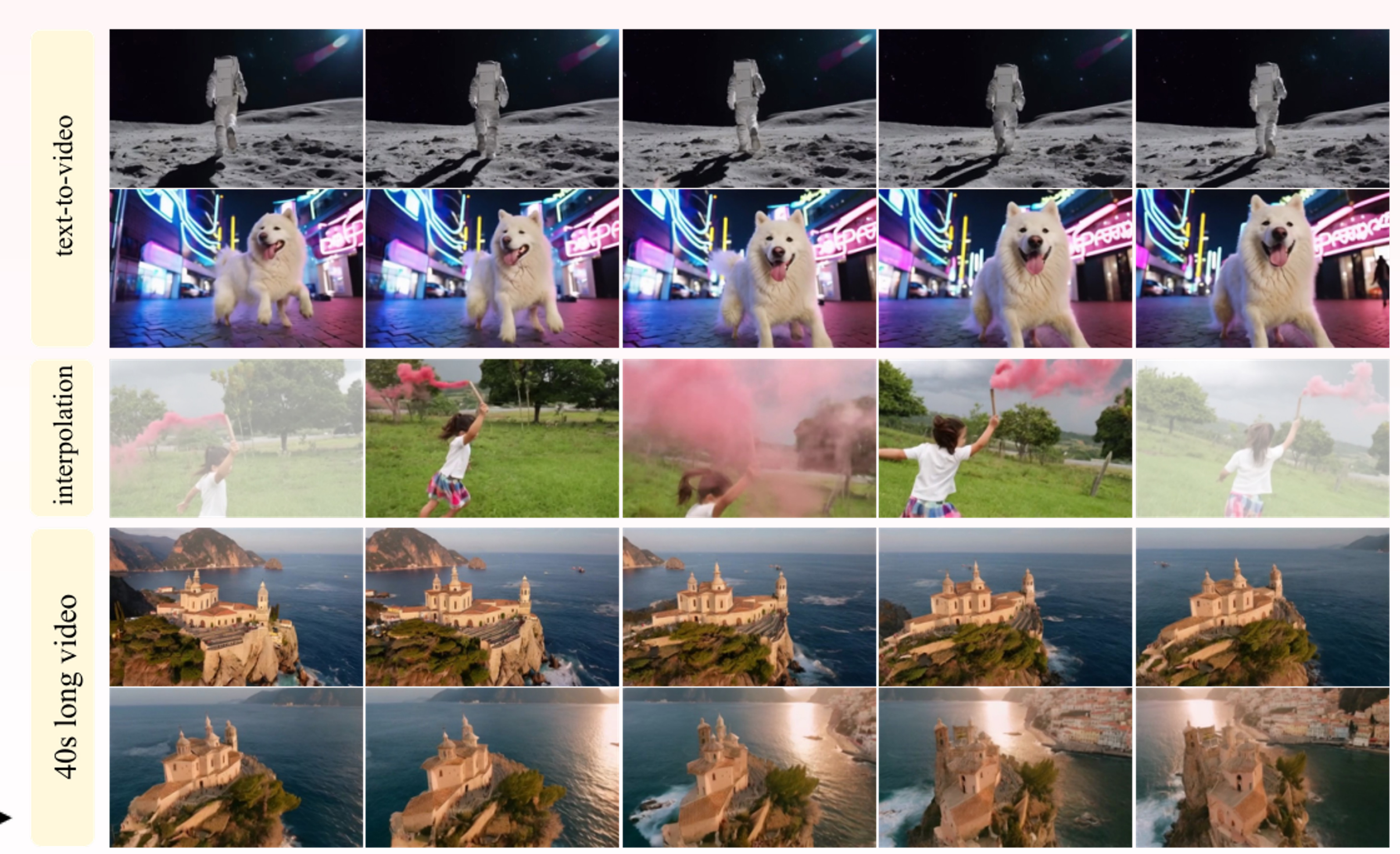
Starting from categorical noise x_0 (left), our framework refines data based on ***token distance*** to get target data x_1 (right), enabling hierarchical structure generation from global semantics to fine details.



Results

URSA rivals Sora-like text-to-video generation models despite using a ***discrete video tokenizer*** and performs on par with the state-of-the-art models in generating high-resolution images.

Model	#params	#videos	Total Score	Quality Score	Semantic Score	Dynamic Degree	Aesthetic Quality	Imaging Quality	Object Class	Multiple Objects	Spatial Relationship	Color	Scene
▼ *Continuous models*													
NOVA	0.6B	20M	80.1	80.4	79.1	20.1	59.4	59.4	92.0	77.5	77.5	87.7	54.1
Vchitect-2.0	2B	134M	81.6	82.5	77.8	58.3	61.5	65.6	87.8	69.4	54.6	86.9	57.5
Pyramid Flow	2B	10M	81.7	84.7	69.6	64.6	63.3	65.0	86.7	50.7	59.5	82.9	43.2
LuminaVideo	2B	12M	83.0	83.9	79.3	67.1	62.3	64.6	91.0	68.3	67.3	90.2	56.1
OpenSoraPlan v1.5	8B	40M	83.0	84.2	78.2	64.4	66.9	68.5	91.9	70.7	80.1	81.8	52.1
OpenSora 2.0	11B	85M	83.6	84.4	80.3	56.4	65.3	65.7	94.6	78.0	76.8	86.3	53.4
MAGI-1	24B	-	81.8	84.7	70.4	72.5	59.3	65.3	84.1	50.6	73.0	87.5	28.9
Step-Video	30B	-	81.8	84.5	71.3	53.1	61.2	70.6	80.6	50.6	71.5	88.3	24.4
CogVideoX1.5	5B	-	82.0	82.7	79.2	56.2	62.1	65.3	83.4	65.3	79.4	88.4	53.3
HunyuanVideo	13B	-	83.2	85.1	75.8	70.8	60.4	67.6	86.1	68.6	68.7	91.6	53.9
Wan2.1	14B	-	83.7	85.6	76.1	65.5	66.1	69.4	86.3	69.6	75.4	88.6	45.8
▼ *Discrete models*													
Lumos-1	3.6B	10M	78.3	79.5	73.5	-	-	58.0	90.1	-	-	82.0	-
Emu3	8B	-	81.0	84.1	68.4	79.3	59.6	62.6	86.2	44.6	68.7	88.3	37.1
URSA	1.7B	24M	82.4	83.4	78.5	81.4	63.1	62.2	93.4	70.6	62.1	90.7	52.3



Model	ModelSpec		DPG-Bench				GenEval						
	#params	#images	Overall	Entity	Attribute	Relation	Overall	Single	Two	Counting	Colors	Position	ColorAttr
▼ *Continuous models*													
SDXL	2.6B	-	74.7	82.4	80.9	86.8	0.55	0.98	0.44	0.39	0.85	0.15	0.23
SD3	2B	-	84.1	91.0	88.8	80.7	0.62	0.98	0.74	0.63	0.67	0.34	0.36
FLUX.1-dev	12B	-	84.9	-	-	-	0.68	0.99	0.85	0.74	0.79	0.21	0.48
NOVA	1.4B	600M	83.0	88.7	86.4	91.9	0.71	0.99	0.91	0.62	0.85	0.33	0.56
SANA-1.5†	4.8B	50M	84.7	-	-	-	0.81	0.99	0.93	0.86	0.84	0.59	0.65
OmniGen2	7B	-	83.6	88.8	90.2	89.4	0.80	1.00	0.95	0.64	0.88	0.55	0.76
Mogao†	7B	-	84.3	90.0	88.3	93.2	0.89	1.00	0.97	0.83	0.93	0.84	0.80
Bagel	14B	-	85.1	90.4	91.3	90.8	0.82	0.99	0.94	0.81	0.88	0.64	0.63
Show-o2†	7B	66M	86.1	91.8	90.0	91.8	0.76	1.00	0.87	0.58	0.92	0.52	0.62
▼ *Discrete models*													
Show-o	1.3B	2B	67.3	75.4	78.0	84.5	0.68	0.98	0.80	0.66	0.84	0.31	0.50
Emu3†	8B	-	81.6	87.2	86.3	90.6	0.66	0.99	0.81	0.42	0.80	0.49	0.45
FUDOKI	1.5B	13M	83.6	89.7	88.1	93.7	0.77	0.96	0.85	0.56	0.88	0.68	0.67
Janus-Pro	7B	72M	84.2	88.9	89.4	89.3	0.80	0.99	0.89	0.59	0.90	0.79	0.66
Meissonic	1B	210M	-	-	-	-	0.54	0.99	0.66	0.42	0.86	0.10	0.22
Lumos-1†	3.6B	60M	-	-	-	-	0.66	0.95	0.80	0.46	0.81	0.48	0.48
Infinity†	2B	-	83.5	-	-	90.8	0.73	0.99	0.85	0.64	0.84	0.49	0.57
URSA (512×320)	1.7B	30M	82.5	88.3	86.4	92.9	0.64	0.99	0.83	0.47	0.83	0.30	0.41
URSA (1024×1024)	1.7B	30M	86.0	91.5	89.6	94.7	0.68	0.99	0.92	0.63	0.86	0.25	0.40
URSA† (1024×1024)	1.7B	30M	-	-	-	-	0.80	1.00	0.92	0.64	0.89	0.67	0.69